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Economic Development Quarterly
2018, Vol. 32(2) 163–176
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DOI: 10.1177/0891242418760981
journals.sagepub.com/home/edq
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Knowledge-Based Regional Economic Development: A Synthetic Review of Knowledge Spillovers, Entrepreneurship, and Entrepreneurial Ecosystems

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Abstract

The geographical nature of knowledge spillovers and entrepreneurship has been well documented in the literature. In most regional or urban studies, these two topics are separately discussed. This research reviews literature at the intersection of knowledge spillovers, entrepreneurship, and regional economic development. The key argument is that entrepreneurship can serve as a mechanism for transmitting knowledge spillovers and accordingly contribute to regional innovation, cluster formation, and economic development. However, the effectiveness of this mechanism depends on various factors in the regional entrepreneurial ecosystem, notably including knowledge bases, absorptive capacity, competition, networks of people, diversity, and culture.

Keywords

entrepreneurship, innovation, entrepreneurial ecosystems, regional development, spin-offs

This review article focuses on knowledge-based regional economic development in which entrepreneurship serves as a mechanism for transmitting knowledge spillovers. Knowledge has long been recognized as a key component of regional economic development. Alfred Marshall (1920) explained industrial clustering by means of a shared labor force with knowledge specialized for a specific industry and knowledge flows between geographically proximate firms. Schumpeter (1934) argued that entrepreneurial activity of combining existing knowledge and appropriating its market value is fundamental to a dynamic capitalist economy. The role of knowledge is also well integrated into mainstream economic growth theories. In the neoclassical growth theory, Solow (1957) considered exogenous technical change (i.e., technical knowledge at the macro level) to be a major predictor of economic performance, especially in the U.S. context. Because of the assumptions in his theory, Solow does not cover how knowledge can be created and further exploited commercially.

Although the economic role of knowledge drew attention a long time ago, it was not until the 1980s that knowledge started to take the central stage in economic growth theories. Two mainstream theories were particularly important to this. The first is the knowledge production function (KPF), which studies the relationship between industrial and university research and development (R&D) input and the innovative output of the private sector (Griliches, 1979; Jaffe, 1986, 1989). KPF is similar to the Cobb–Douglas production function used by Solow (1957) but focuses on the output of innovations, not products or services. The second is the endogenous growth theory (EGT) developed by Romer (1990). Instead of considering knowledge a nonrival, nonexcludable public good (Arrow, 1962), Romer assumed knowledge nonrival but partially excludable (e.g., via the patent system). This explains why private businesses have incentives to invest in R&D. According to EGT, knowledge drives economic development in the long run.

Except for the work of Jaffe (1986, 1989), the above mentioned mainstream economic growth theorists do not directly discuss the geographical nature of knowledge. Despite being considered nonrival and nonexcludable (or partially excludable in EGT), knowledge bases vary across regional economies (Asheim & Coenen, 2005; Audretsch & Feldman, 1996; Feldman & Audretsch, 1999). For instance, the intensity of engineering knowledge is much higher in Silicon Valley than in most other U.S. regions, and the biomedical knowledge intensity is significantly higher in notable U.S. college towns such as Durham (home to the University of North Carolina and Duke University), Gainesville (home to the University

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of Florida), and Iowa City (home to the University of Iowa; Qian, 2017). In explaining the stickiness of regional knowledge bases, Asheim and Coenen (2005) stress the role of labor market history. The geographically varying knowledge bases suggest the importance of a regional perspective in applying KPF and EGT.

Knowledge spillovers are discussed in both KPF and EGT. KPF studies knowledge spillovers from universities to businesses. EGT assumes knowledge spills over from businesses to society, which improves the technical level of the entire economy. From a regional perspective, numerous empirical studies show that such spillover effects are geographically bounded (Anselin, Varga, & Acs, 1997; Audretsch & Feldman, 1996; Autant-Bernard & LeSage, 2011; Jaffe, 1986, 1989; Jaffe, Trajtenberg, & Henderson, 1993; Peri, 2005). Despite the recognition of the importance of knowledge spillovers, neither KPF nor EGT has explored the mechanisms of how knowledge spills over between economic agents (Acs, Audretsch, Braunerhjelm, & Carlsson, 2009). In regional economic development, this is a critical question to understand the connections between locally produced/endowed knowledge and regional economic growth. As noted by Braunerhjelm, Acs, Audretsch, and Carlsson (2010), there is a "missing link" in EGT that casts shadow over the role of knowledge in long-term economic growth.

A group of scholars (Acs & Armington, 2006; Acs & Plummer, 2005; Acs, Audretsch, Braunerhjelm, & Carlsson, 2009; Audretsch, 1995; Audretsch & Keilbach, 2008; Audretsch, Keilbach, & Lehmann, 2006; Audretsch & Lehmann, 2005; Qian, Acs, & Stough, 2013) have addressed this missing link with the *knowledge spillover theory of entrepreneurship* (KSTE). This theory, in short, considers entrepreneurship a mechanism of transmitting knowledge spillovers. Research on KSTE has flourished in recent years, as noted in the review piece by Ghio, Guerini, Lehmann, and Rossi-Lamastra (2015), much in the context of regional economic development and often with some discussion on policy (e.g., Acs & Armington, 2006; Audretsch & Lehmann, 2005; Qian et al., 2013). As KSTE is a relatively theoretical economic model, existing work has fallen short when describing the theory and its policy implications to economic development policy makers and practitioners. Considering this disconnection between research and practice, the objective of this research is to provide a comprehensive review of KSTE in regional economic development. Instead of a critique approach that would typically seek for the weaknesses of an existing theory and further lay out future research directions, this review adopts a synthesis approach, coherently presenting the fragmented KSTE research to economic development scholars who are not familiar with KSTE, or economic development policy makers and practitioners who are interested in knowledge-based economic development strategies.

This synthetic review makes three contributions to the literature. First, it comprehensively reviews the literature on entrepreneurship as one mechanism of transmitting knowledge spillovers in the context of regional economic development. Existing reviews on KSTE (Acs et al., 2013; Ghio et al., 2015) have not sufficiently addressed the regional context or economic geography. Second, it brings together two streams of literature that are closely related but, strikingly, lack dialogues so far: knowledge spillover entrepreneurship and spin-offs. This effort also leads to the discovery of a new mechanism by which knowledge spillover entrepreneurship contributes to regional economic development; that is, through building industrial clusters. Third, it introduces the entrepreneurial ecosystems framework to discuss regional factors that are particularly important to knowledge spillover entrepreneurship, which sheds light on economic development policy. The entrepreneurial ecosystems approach has recently gained popularity (Feld, 2012; Isenberg, 2011; Qian et al., 2013; Spigel, 2017; Stam, 2015). Existing research has not customized entrepreneurial ecosystems for knowledge spillover entrepreneurship.

The review is organized as follows. After this introduction, the second section reviews some literature on knowledge spillovers in economic geography. It is worth noting that the purpose of this section is not to provide a comprehensive review of geographically mediated knowledge spillovers, which has been done by others (e.g., Döring & Schnellenbach, 2006), but rather to provide conceptual and theoretical backgrounds for the mechanism of knowledge spillovers explored later. The next section discusses entrepreneurship as one mechanism of transmitting knowledge spillovers, while the fourth section connects KSTE with the literature on spin-offs. The fifth section systematically explores the regional context in which knowledge spillover entrepreneurship can be effectively exercised. The last section concludes by briefly discussing economic development policy that facilitates knowledge spillover entrepreneurship.

The Economic Geography of Knowledge Spillovers

Geography offers an important perspective to understand the development and diffusion of human knowledge. Howells (2002) noted five ways by which the geographical context matters to knowledge:

1. The development of human knowledge is shaped by social, economic, cultural, and cognitive environments that are geographically specific.
2. The development of human knowledge also depends on social interaction in which distance and place can make a difference.
3. Acquiring external information, similarly, is affected by distance and place.
4. The learning process is socially, economically, and territorially constrained.
5. Interpreting information requires contextually embedded knowledge shaped by experience and geography.

Except for the first one, these explanations on the importance of geography to knowledge involve knowledge flows, which include knowledge spillovers and transfers.

Knowledge spillovers are different from *knowledge transfers*. Both types of knowledge flows play an important role in economic development, but the former appears to be more important at the macro, regional, or societal level. By definition, knowledge spillovers do not involve compensation (or full market-rate compensation) to knowledge "holders" from knowledge "recipients" when the flows happen (Agarwal, Audretsch, & Sarkar, 2010). This is particularly important at the macro level, as it implies that the societal knowledge level can increase without incurring much additional learning or R&D cost. However, knowledge spillovers do not necessarily result in higher levels of societal knowledge (Feldman, 2014). It is also likely that strong knowledge spillovers would discourage new R&D investment and encourage more economic agents to wait to be free riders. This is supported by the inverted U-shaped relationship between patent protection and economic growth reported by Acs and Sanders (2012). Knowledge transfers, on the other hand, require market-rate compensation in obtaining knowledge from others (individuals or firms). In such cases, both the benefits (of appropriating the market value of new knowledge) and the costs (of developing new knowledge in R&D) are transferred from one party to another. Unlike knowledge spillovers, it is common that only one party (i.e., the ultimate owner) can commercially exploit the knowledge when intellectual property protection is in place. In practice, it is difficult to separate knowledge spillovers from knowledge transfers due to the improbable task of deciding the market value of new knowledge *prior to* its commercialization (Agarwal et al., 2010). The ambiguous boundary between knowledge spillovers and knowledge transfers has implications on the relationship between knowledge spillovers and entrepreneurship, which will be discussed later when the KSTE is introduced.

Knowledge spillovers between firms can occur by means of human capital mobility or face-to-face communication between skilled employees from different firms (Döring & Schnellenbach, 2006). In the former approach, human capital is usually defined as knowledge embodied in people (Schultz, 1961). When skilled workers change jobs, they spread the knowledge from their previous employers to the new ones. Breschi and Lissoni (2001, p.991), however, warned that these skilled workers may simply "shift" the knowledge embodied in them instead of spreading it when such knowledge is not made known to their former employers/colleagues. Therefore, human capital mobility does not always lead to knowledge spillovers. The latter approach (i.e., face-to-face interaction) is built on the *codified* versus *tacit* typology of knowledge (Polanyi, 1967). Tacit knowledge is difficult to transmit without face-to-face interaction because of communication difficulties and/or lack of self-awareness (Gertler, 2003).

Geography plays a role in both human capital mobility and face-to-face interaction, and therefore affects knowledge spillovers. In terms of human capital mobility, it is much more likely that skilled workers switch to new jobs within the same region. The opportunity costs of relocation to a new region are usually high, not only because of the high transaction costs of selling/buying homes but also due to the risk of losing established local networks. In term of face-to-face interaction, geography facilitates the spillover of tacit knowledge, as geographically proximate individuals are more likely to meet, either purposefully or serendipitously. Moreover, face-to-face knowledge spillovers are easier between people from the same region when the knowledge is contextually specific to the region (Gertler, 2003). Numerous studies have demonstrated that knowledge spillovers are localized (Anselin et al., 1997; Audretsch & Feldman, 1996; Autant-Bernard & LeSage, 2011; Jaffe, 1986, 1989; Jaffe et al., 1993; Peri, 2005). Anselin et al. (1997), for example, found that knowledge spillovers from universities are no longer significant beyond 50 miles.

Despite all the empirical evidence from these geographical studies, it is worth noting that the role of geography should not be exaggerated. The idea of geographically bounded knowledge spillovers is criticized even by economic geographers. For example, Boschma (2005, p.61) argued that geographical proximity is "neither a necessary nor a sufficient condition for learning"; other types of proximities (i.e., cognitive, organizational, social, and institutional) also matter.

The extent to which geography mediates knowledge spillovers, according to Asheim, Boschma, and Cooke (2011), depends on the knowledge bases of regional economies. Asheim and his colleagues (Asheim, Coenen, & Vang, 2007; Asheim et al., 2011; Asheim & Hansen, 2009) identified three types of knowledge bases: analytical (science based), synthetic (engineering based), and symbolic (art based). Analytical knowledge is mostly codified and therefore its spillovers rely less on face-to-face interaction. In contrast, synthetic knowledge addresses the application or combination of existing knowledge, in which the flows of tacit knowledge (or know-how) become important. Symbolic knowledge involves aesthetic or artistic characteristics of products, requiring not only know-how but also know-who (i.e., identifying the person who can perform a very specific task). Therefore, geographical proximity and geographically mediated cognitive, social, and economic circumstances may play important roles in commercializing synthetic knowledge and symbolic knowledge. The empirical work by Qian (2017) confirmed Asheim's conceptual framework, which reports positive associations of engineering knowledge (i.e., synthetic) and arts knowledge (i.e., symbolic), but not biomedical knowledge (i.e., analytical), with innovative activity in cities. In addition, Qian's typology also featured other types of knowledge bases, notably the management knowledge of cities. This corresponds to management know-how emphasized by Bhidé (2008) and Malecki (2010), which is tacit in nature and requires face-to-face interaction. Qian (2017) confirmed that the innovation impact of management knowledge is also geographically bounded.

Also related to economic geography, knowledge spillovers are widely studied in the context of regional industrial structures. Glaeser, Kallal, Scheinkman, and Shleifer (1992) conceptualized the spillover debate between Marshall–Arrow–Romer (MAR) externalities and Jacobian externalities. On one side, Marshall (1920), Arrow (1962), and Romer (1986) argued that knowledge spillovers are facilitated by the spatial concentration of similar firms (i.e., specialization). On the other side, Jacobs (1969) suggested that spillovers across different types of firms (i.e., diversity) are critical to the vitality of urban economies. Several empirical studies have aimed to explain whether specialization or diversity matters. A review by Beaudry and Schiffauerova (2009) based on 67 empirical studies showed that about 70% support MAR externalities and 75% find evidence of Jacobian externalities. Therefore, there has been no consensus over how regional or local industrial structures may affect localized knowledge spillovers. Some recent work refines Jacobian externalities, revealing the importance of *related* variety (vs. unrelated variety) to knowledge spillovers (Asheim et al., 2011; Boschma & Iammarino, 2009; Frenken, Van Oort, & Verburg, 2007). Similarly, Delgado, Porter, and Stern (2014) reported the contribution of clusters (defined as a group of interconnected industries in a region) to knowledge spillovers.

Knowledge Spillover Entrepreneurship and Its Role in Innovation of Cities

Similar to knowledge and knowledge spillovers, entrepreneurship is also widely studied in economic geography (Ács, Szerb, Ortega-Argilés, Aidis, & Coduras, 2015; Belitski & Desai, 2016; Bosma & Sternberg, 2014). It needs to be first noted that entrepreneurship is a multidimensional concept. For instance, Schumpeter (1934) considered entrepreneurship the same as innovation and defined entrepreneurs as those carrying out innovative activity. In the management literature, entrepreneurship usually refers to the process of discovering and exploiting market opportunities as well as the traits, personalities, and behavior of entrepreneurs (e.g., Casson, 1982; Shane & Venkataraman, 2000). Another perspective of entrepreneurship is from new and dynamic businesses (Drucker, 1985) that have great economic impacts. The approach to entrepreneurship in this research is on new business formation by entrepreneurs, especially the creation of knowledge-driven, growth-oriented firms that contribute significantly to regional economic development.

The impact of entrepreneurship or new firm formation on regional development has been well documented in the literature. Often studied is the job creation effects of new businesses. Haltiwanger, Jarmin, and Miranda (2013) found that new businesses account for all net job growth in the U.S. economy. Acs and Armington (2006) confirmed new firm formation as a driving force of the employment growth in U.S. metropolitan regions. New firm formation is also associated with market efficiency, as it drives the economy toward equilibriums through a more efficient use of market resources (Kirzner, 1997). This review focuses on another role of entrepreneurship in regional development; that is, by way of transmitting knowledge spillovers that underpin an innovative economy. Knowledge spillovers are critically important to innovation (typically understood as new products, services, or production processes) in cities or regions (Anselin et al., 1997; Feldman & Audretsch, 1999; Lucas, 1988). By transmitting knowledge spillovers, entrepreneurs introduce new products or services into the market based on knowledge that would otherwise be left uncommercialized (Acs et al., 2009). Entrepreneurship therefore contributes to innovation and ultimately to regional economic development.

To understand the role of entrepreneurship in economic development by means of knowledge spillovers, we return to economic growth theories discussed in the first section. As noted, there is a missing link in EGT between knowledge and economic growth through innovative activity (Braunerhjelm et al., 2010). Indeed, not all new knowledge has economic value and accordingly can be commercialized (Arrow, 1962). Even for knowledge with potential for commercial use, the appropriation of its market value is not warrantied. Carlsson, Acs, Audretsch, and Braunerhjelm (2009, p.1196) used the "knowledge filter" concept to describe this phenomenon, defined as "the barriers to converting research into commercialized knowledge." The knowledge filter exists because of both the nature of knowledge itself and the context in which knowledge can potentially be brought into the market. On one hand, Arrow (1962) characterized knowledge (as an input to economic output) by high levels of uncertainty, asymmetries, and transaction costs (cited in Audretsch & Lehmann, 2005), making its commercialization very difficult. On the other hand, there are organizational and institutional barriers to the realization of the market value of new knowledge (Carlsson et al., 2009), which applies particularly to the knowledge created in universities (Qian & Yao, 2017).

For commercially viable new knowledge, there are three primary ways of appropriating its market value: (1) incumbent firms introduce new products based on new technologies developed in their own research labs; (2) new products arise from technology transfers between different organizations (e.g., from a university to a firm); and (3) knowledge spills over between economic agents, and the recipient better recognizes its market value and acts faster to commercialize it. The focus here is on number 3. As Acs et al. (2009) and Audretsch (1995) have noted, the mechanism of knowledge spillovers as one way of commercializing new knowledge is not well studied in KPF and EGT. In the previous section, it was discussed that knowledge spillovers can occur when skilled workers move between companies or those from different firms interact with one another, either purposefully or serendipitously (Döring & Schnellenbach, 2006). These two scenarios, both mediated by geography, only involve knowledge flows by definition and do not necessarily lead to knowledge commercialization. Knowledge spillovers by way of entrepreneurship, as discussed next, directly address the commercialization process.

To better understand how spillovers lead to knowledge commercialization, Acs, Audretsch, and their collaborators developed the KSTE (Acs & Armington, 2006; Acs et al., 2009; Acs & Plummer, 2005; Audretsch, 1995; Audretsch et al., 2006; Audretsch & Keilbach, 2008; Audretsch & Lehmann, 2005; Qian et al., 2013). They consider entrepreneurship, or new firm formation, as one of the key mechanisms of transmitting knowledge spillovers. New knowledge under this theory is one major source of entrepreneurial opportunities. Knowledge spillover entrepreneurship typically involves scientific entrepreneurs. For instance, a scientist in an incumbent firm develops a new idea that has potential market value, but the firm's decision makers may choose not to commercialize it when compared with other technological opportunities or simply because of inertia. The scientist who initially developed the new idea may have a higher expectation of its market potential than his/her employer. Knowledge spillover entrepreneurship occurs when the scientist decides to leave his/her employer, start a new business, and then commercialize it as an entrepreneur. In some cases, the scientist entrepreneur may have to compensate his/her employer to commercially exploit the idea (due to institutional intellectual property protection). However, the employer, with the decision not to commercialize it, has not fully valued the new idea and is likely to be undercompensated compared with its real market value. By definition (Agarwal et al., 2010), this undercompensated knowledge flow between the incumbent firm and the new firm is still a knowledge spillover. In the case of university knowledge, the entrepreneurial activity of faculty, research staff, or students follows a similar logic but with different reasons. Universities do not directly commercialize new technologies they own or new ideas that originate from them. When technology transfer to incumbent firms is not a viable option, a faculty member may start a new firm to exploit a new technology or idea she/he generates at the university. In many cases, these academic entrepreneurs do not keep their faculty jobs, either voluntarily because of pursuing entrepreneurship as the more attractive career, or forced to do so in accordance with university rules (Qian & Yao, 2017).

Built on the work of Cohen and Levinthal (1990), Qian and Acs (2013, p. 191) advanced KSTE by highlighting entrepreneurial absorptive capacity (i.e., "the ability of an entrepreneur to understand new knowledge, recognize its value, and subsequently commercialize it by creating a firm"). They argued that the occurrence of knowledge spillover entrepreneurship (i.e., entrepreneurship as one mechanism of knowledge commercialization via spillovers) depends on both new knowledge creation and entrepreneurial absorptive capacity. The latter consists of two components: technical and business skills. By introducing entrepreneurial absorptive capacity, Qian and Acs also extend the application of KSTE to the behavior of noninventor entrepreneurs, which leads to knowledge spillovers not only between organizations but also between individuals. For example, an entrepreneur not affiliated with a university may exploit the market value of a new idea that originates from the university. In this case, the entrepreneur himself/herself is not the inventor, but has sufficient technical background to understand the new idea and discover its potential market value when communicating with the university research staff. Qian and Acs also addressed the importance of human capital to both new knowledge generation and absorptive capacity building, and ultimately to knowledge spillover entrepreneurship. Empirically, Qian (2016) found both direct and moderating effects of various labor market skills (as measures of human capital) on knowledge spillover entrepreneurship in U.S. cities. These skills, including cognitive, social, technological, problem solving, systems, and resources management, particularly moderate the entrepreneurial process of commercializing university knowledge (vs. industry knowledge).

Combined with the geographical nature of localized knowledge spillovers, KSTE has two major implications for regional economic development. First, ceteris paribus, a higher level of local knowledge creation leads to a higher level of start-up activity, as the former gives rise to more opportunities for entrepreneurs to discover and exploit. Therefore, investing in R&D or knowledge creation may contribute to a more entrepreneurial regional economy. Many empirical studies have provided evidence on this positive relationship between knowledge and new firm formation in regional economies (Audretsch, Bönte, & Keilbach, 2008; Audretsch & Lehmann, 2005; Plummer & Acs, 2014; Qian & Acs, 2013; Tsvetkova, 2015). It should be noted that the magnitude of this relationship depends on the extent to which local incumbent firms have the culture of encouraging their knowledge workers to develop new products within the firm, which is typically termed "intrapreneurship." It also depends on the nature of a new product or service: If it is sold directly to customers but not businesses, knowledge spillover entrepreneurship is more likely to happen than intrapreneurship in pursuing the market opportunity (Parker, 2011). At the macro level, Bosma, Stam, and Wennekers (2010) found a negative correlation between intrapreneurship and entrepreneurship. Therefore, if the regional corporate culture is friendlier to the internal exploitation of new opportunities, increased regional knowledge will likely have a smaller positive impact on knowledge spillover entrepreneurship. The relationship, albeit weaker, is still expected to be positive, as the intrapreneurship-friendly incumbents are unlikely to commercialize all the new knowledge and entrepreneurs still have a good chance to commercialize part of it. Moreover, Bosma et al. (2010) reported a higher probability of starting new firms by intrapreneurs than other employees.

The second implication of KSTE for regional development is that, when there is a shortage in the supply of entrepreneurs or entrepreneurial skills, locally created new knowledge is less likely to be locally commercialized. Therefore, new knowledge does not necessarily lead to regional economic growth as EGT has suggested. This is the problem in many American college towns where knowledge production is high but entrepreneurial activity is low (Qian & Yao, 2017). Localized entrepreneurial activity, according to KSTE, addresses this "missing link" (Braunerhjelm et al., 2010) and allows more knowledge to be locally commercialized. New products or processes introduced by entrepreneurs from knowledge spillovers boost innovation (Acs et al., 2009).

Spin-Offs: Why Knowledge Spillover Entrepreneurship Matters for Cluster Formation

To further explore the mechanism of knowledge spillover entrepreneurship, it is useful to refer to the literature on spin-offs. Spin-offs are firms founded by former employees of incumbent firms (or of universities in the case of academic spin-offs). Knowledge spillover entrepreneurship can to a large extent be considered as innovation-driven spin-offs, but surprisingly, KSTE scholars have rarely connected their theory with the spin-offs literature. This disconnection perhaps results from the mismatched parts between knowledge spillover entrepreneurship and spin-offs. On one hand, spin-off activity often does not involve innovations; employee start-ups can simply sell the same or similar products as their parents (Klepper, 2001). In this scenario, spin-offs are, at best, loosely connected to knowledge spillover entrepreneurship that addresses the commercialization of new knowledge. On the other hand, knowledge spillover entrepreneurship may not necessarily involve spin-offs; an outside entrepreneur can commercialize new knowledge created in a university or in an incumbent firm for which it is difficult to establish a property right (Qian & Acs, 2013). In this scenario, the entrepreneur's new firm is not a spin-off of the university or the incumbent. Despite these mismatched parts, innovation-driven spin-off activity is well matched with knowledge spillover entrepreneurship involving spin-offs, which represents the overlapped part between spin-off activity and knowledge spillover entrepreneurship.

Klepper (2001), in a review piece, discussed the agency theories of spin-offs and pointed out "the essential idea of these [agency] theories is that spin-offs develop innovations that originate in efforts by their parents" (p. 661). These innovations are not pursued by the parent firms "because of various contracting problems that lead employees to develop the innovations in their own firms" (p. 661). These agency theories of spin-offs explain the same phenomenon with knowledge spillover entrepreneurship. Innovation-based spin-offs described in agency theories are prevailing in the disk drive industry, but less so in other industries such as semiconductors and lasers (Klepper, 2001), suggesting industry variations in knowledge spillover entrepreneurship. Even within the same industry, spin-offs are more likely to pursue some kinds of innovation over others. As demonstrated by the organizational capacities theories that Klepper (2001) has also reviewed, spin-offs have comparative advantage in pursuing innovations that are "architectural, competence-destroying, opening new submarkets, or ones involving modest capital and vague business plans" (p. 646). Regardless of industry, there is strong evidence showing that "firms with richer and broader knowledge spawned more spin-offs" (Klepper, 2001, p. 657). This finding supports KSTE at the firm level.

Having connected knowledge spillover entrepreneurship with innovation-driven spin-offs, the well-established literature on the relationship between spin-offs and industrial clustering provides new insight into the role of knowledge spillover entrepreneurship in regional economic development. As discussed earlier, MAR externalities suggest that industrial clustering contributes to knowledge spillovers and therefore to knowledge spillover entrepreneurship. Challenging this traditional thinking, Klepper and his collaborators (Buenstorf & Klepper, 2009; Golman & Klepper, 2016; Klepper, 2001, 2007, 2010, 2011) argued that it is localized spin-off activity, and not benefits from agglomeration economies, that leads to industrial clustering. Their investigations on different industries (such as semiconductors in Silicon Valley, tires in Akron, and automobiles in Detroit) revealed similar patterns of industrial clustering: One superior firm started somewhere, often serendipitously, and spin-offs from this firm grew the industry into a strong cluster in the same place. This evolutionary process is facilitated by the key finding of the heritage theory that superior firms have more competent spin-offs, which is discussed widely in Klepper's work. Spin-offs that perform well often inherit organizational competency from their parents, and their founders have gained "in-depth knowledge about technology, production processes and the market in which they compete" through earlier experience (Buenstorf & Klepper, 2009, p. 730). In Silicon Valley, for example, 93% of documented entrants in its semiconductor industry were spin-offs and 82% of these spin-offs could be traced to the notable first entrant, Fairchild Semiconductors (Klepper, 2011). It has not been well studied as to why employee entrepreneurs often locate spin-offs near their parents, but it is logical to consider factors such as high relocation costs, which include the opportunity cost of losing local networks (Dahl & Sorenson, 2012) and factors that benefit their parents, such as local specialized suppliers and services (Boschma, 2015).

In addition to its role in innovation as discussed in the previous section, the literature on spin-offs clearly suggests another role of knowledge spillover entrepreneurship in regional economic development: driving cluster formation. This role has been well demonstrated in the evolutions of the semiconductor cluster in Silicon Valley and other clusters in Akron, Detroit, and Washington, D.C. (Feldman, Francis, & Bercovitz, 2005; Klepper, 2011). Furthermore, clusters have many benefits to productivity growth and constitute the competitive advantage of regions (Porter, 1998).

Regardless of the mechanism (i.e., via innovation suggested by the KSTE literature or cluster formation suggested by the spin-offs literature), the overall effect of knowledge spillover entrepreneurship on regional development has been empirically tested in various geographical contexts. For instance, Audretsch and Keilbach (2008) confirmed the contribution of knowledge spillover entrepreneurship (measured by entrepreneurial activity in high technology industries and ICT industries) to regional output based on an analysis of 440 German counties. Their empirical test is theoretically grounded in KSTE. Using U.S. metropolitan statistical areas as the geographic unit, Qian and Jung (2017) and Tsvetkova (2015) reported a positive, significant moderating effect of knowledge-based entrepreneurship on the relationship between knowledge and regional economic performance. Specifically, Qian and Jung (2017) found that high technology entrepreneurship strengthens the positive association between knowledge measured by patents and metropolitan productivity measured by GDP per worker. The complete model in Tsvetkova (2015) exhibited a negative association between knowledge measured by patents and economic performance measured by the number of expanding firms, job creation by expanding firms, and job creation by new firms in the knowledge-intensive professional, scientific, and technical services industry. This relationship will be significantly mitigated with a higher level of start-up activity in this industry because of the positive moderating effect of entrepreneurship within the industry. Based on county-level data in Colorado, Acs and Plummer (2005) similarly provided the evidence of a positive moderating effect of high technology entrepreneurship on the relationship between knowledge measured by patents and economic performance measured by personal income growth. It must be pointed out that these empirical studies have one shared weakness: Knowledge spillover entrepreneurship is not directly measured. This is because regional-level data cannot separate knowledge spillover entrepreneurship from other types of entrepreneurship. Scholars often use entrepreneurship in knowledge-intensive sectors as a proxy, exemplified by all the articles cited in this paragraph (Acs & Plummer, 2005; Audretsch & Keilbach, 2008; Qian & Jung, 2017; Tsvetkova, 2015).

For regions that adopt an innovation- or cluster-based economic development strategy, it will be useful to create an environment that encourages knowledge spillover entrepreneurship. Next, we discuss the regional context that affects knowledge spillover entrepreneurship.

Context Matters: An Ecosystems Approach to Knowledge Spillover Entrepreneurship

Knowledge spillover entrepreneurship, as discussed earlier, plays a significant role in innovation, cluster formation, and ultimately regional economic development. To strengthen it, simply investing in human capital, as suggested by Qian and Acs (2013), is not sufficient. The regional context in which entrepreneurs exploit the market value of new knowledge matters. To this point, a recent conceptual framework—regional entrepreneurial ecosystems—offers some insight into building strong knowledge spillover entrepreneurship for the regional economy. An ecosystems approach to entrepreneurship addresses a holistic view of important factors that facilitate local entrepreneurial activity (Qian et al., 2013), especially for innovative or high-growth start-ups (Stam, 2015). The interaction between individual entrepreneurs and their social and economic environments lies at the core of entrepreneurial ecosystem studies. Isenberg (2011) summarized the environmental factors as policy, markets, finance, human capital, culture, and supports. The work by Spigel (2017) includes a longer list of factors: culture, histories, networks, capital, deal makers, mentors, talent, universities, supports, policy, and openness. Stam (2015) divided the components of entrepreneurial ecosystems into two categories: framework conditions including institutions, culture, infrastructure, and demand; and systemic conditions including networks, leadership, finance, talent, knowledge, and support services. Regardless of the factors included, the ecosystems scholars adopt a holistic approach and consider all these factors at the same time. The interactions of different factors are also of interest in entrepreneurial ecosystems studies.

Existing studies on entrepreneurial ecosystems have been generalized and not customized for knowledge spillover entrepreneurship. As one of the key contributions of this review, this section identifies regional environmental factors that are particularly important to knowledge spillover entrepreneurship. These factors include knowledge bases, competition, networks, diversity, and culture. The focus on these five factors does NOT mean that other factors (e.g., available capital, demand, infrastructure, and leadership) have no effect on knowledge spillover entrepreneurship; the ecosystems literature has suggested they do. They are not discussed here because they are important to general entrepreneurial activity but not *particularly* to knowledge spillover entrepreneurship, or at least have not been examined in the KSTE framework. The five factors of focus are chosen based on a thorough review on the literature; they are the ones that have been theoretically discussed and empirically studied in the KSTE framework or its closely related innovation-driven spin-offs literature.

First, knowledge spillover entrepreneurship depends on the *knowledge bases* of regions or cities. Knowledge is not born equal when it comes to commercial potential. Some knowledge bases are more conducive to entrepreneurial opportunities than others. This is indirectly supported by industry variations in innovation-driven spin-offs (Klepper, 2001) and the fact that different industries have different knowledge bases. KSTE typically involves commercializing R&D output by technology entrepreneurs. R&D output reinforces the analytical or synthetic knowledge base identified by Asheim et al. (2007). Compared with synthetic knowledge, however, the science-based analytical knowledge seems further away from prototypes or new products and is less constrained by geographic boundaries (Asheim et al., 2007). Qian (2017) conducted a systematic study on the association between different knowledge bases and high technology entrepreneurship (as a proxy for knowledge spillover entrepreneurship) in U.S. cities. He found that the engineering knowledge base (consistent with Asheim's synthetic knowledge) and the management knowledge base are positively associated with high technology start-up activity, in contrast to the negative associations from the biomedical knowledge base (consistent with Asheim's analytical knowledge) and the agricultural knowledge base. Considering the empirical evidence from Qian (2017), regions or cities smartly specialized in the engineering and/or management knowledge base have more entrepreneurial opportunities to exploit.

Second, a regional environment encouraging healthy *competition* contributes to knowledge spillover entrepreneurship. Competition among similar firms in a cluster creates peer pressure for businesses to innovate as means to survive and grow (Porter, 1998). This leads to additional knowledge created by incumbent businesses in the region, and therefore creates more knowledge-based entrepreneurial opportunities that can spill over into new firms (Plummer & Acs, 2014). Large incumbent firms with strong R&D capacity are of particular importance in generating new knowledge, but they might deter start-up activity and the diversification of the regional economy due to their monopoly or oligopoly power, as experienced in Detroit (Chinitz, 1961). However, it is also possible that these anchor firms serve as a facilitator of a more diverse economy, such as the role Boeing played in Seattle (Markusen, 1996). Plummer and Acs (2014) theorized this in a competition model of knowledge spillover entrepreneurship. They argued that, when incumbents and entrepreneurs compete for commercializing new knowledge, knowledge spillover entrepreneurship will be negatively affected. Entrepreneurs are clearly in a disadvantageous position to compete for knowledge created and owned by large incumbent firms. In contrast, given the third-party ownership, entrepreneurs are in a much better position to compete for knowledge created in universities. Eventually, it depends on whether the positive effect of localized competition (i.e., knowledge creation) outweighs its negative effect (i.e., the share of knowledge opportunities exploited by entrepreneurs). Empirical evidence (Audretsch, Dohse, & Niebuhr, 2010; Qian, 2013, 2017) has clearly shown that the overall competition effect (typically measured by the reverse of the average firm size or the share of small firms) is good for knowledge spillover entrepreneurship.

Third, of particular importance to knowledge spillover entrepreneurship is the localized *networks of individuals* that can best define an entrepreneurial ecosystem and distinguish it from other similar conceptual frameworks, such as innovation systems. Students of innovation systems do address networks but focus primarily on the networks of organizations, especially firms (Edquist, 2005). In the case of knowledge spillover entrepreneurship, the discovery and assessment of new knowledge as entrepreneurial opportunities occur before a new firm (i.e., an organization) is even created; therefore, the focus on the networks of organizations would not be appropriate in entrepreneurial ecosystems. After all, entrepreneurs are individuals. A series of studies (Mayer, 2012; Motoyama, Danley, Bell-Masterson, Maxwell, & Morelix, 2013; Motoyama, Konczal, Bell-Masterson, & Morelix, 2014; Motoyama & Watkins, 2014) have shown the power of personal networks in the ecosystem, both between entrepreneurs themselves and between entrepreneurs and their supporters such as mentors, angel investors, venture capitalists, and brokers or deal makers. Similarly, practitioners (Feld, 2012) have stated the importance of different kinds of meet-up events as one way to build and strengthen the networks in the local technological entrepreneurial ecosystem. In knowledge spillover entrepreneurship, we often see entrepreneurs create new firms to commercialize others' knowledge. In this scenario, it is essential to build connections between knowledge creators and entrepreneurs, so that new knowledge-based entrepreneurial opportunities are exposed to potential commercialization. In the case of scientific or academic entrepreneurs who create knowledge themselves, interpersonal networks between them and business-oriented mentors and technology investors are critical. The importance of such networks to university spin-offs is well recognized in the literature (Rasmussen, Mosey, & Wright, 2015; Van Burg, Romme, Gilsing, & Reymen, 2008; Walter, Auer, & Ritter, 2006). In practice, meet-up events, entrepreneurship celebrations or expo events, coworking spaces, and other networking opportunities that bring scientists, engineers, entrepreneurs, and technology investors together play important roles in facilitating knowledge spillover entrepreneurship.

Fourth, *diversity* proves to be a contributor to knowledge spillover entrepreneurship. Diversity is twofold here, including both *diversity of related industries* and *diversity of people*. Although the debate over whether industrial specialization or diversity provides greater benefits to knowledge spillovers has been unresolved (Beaudry & Schiffauerova, 2009), there has been ample evidence showing that knowledge spillovers prevail across technologically related (vs. unrelated) industries (Asheim et al., 2011; Boschma & Iammarino, 2009;

Frenken et al., 2007). These related industries have interconnected but not the same knowledge bases, a structure encouraging effective learning (Nooteboom, Van Haverbeke, Duysters, Gilsing, & Van den Oord, 2007). Accordingly, knowledge spillover entrepreneurship is expected to be stronger in a region with a higher level of diversity among related industries. An employee in one industry can use his/her new knowledge to develop a new product belonging to another related industry, the urban innovation phenomenon described by Jane Jacobs (1969). When this is executed in a new firm, knowledge spillover entrepreneurship occurs. Baltzopoulos, Braunerhjelm, and Tikoudis (2016), Bishop (2012), Delgado, Porter, and Stern (2010), and Guo, He, and Li (2016) found direct evidence that diversity of technologically related industries is positively associated with new firm formation in regional economies. Among these studies, Bishop explicitly highlighted KSTE as the theoretical foundation for his empirical test.

Besides diversity of related industries, diversity of people also facilitates knowledge spillover entrepreneurship. This results from the asymmetric nature of knowledge (Arrow, 1962) that leads to different perceptions and valuations of the same piece of knowledge by people with different backgrounds and perspectives of thinking (Audretsch et al., 2010; Qian, 2013). As Audretsch et al. (2010, p.58) state, “The more different kinds of people evaluate any given idea, the higher will be the probability that one of these persons will arrive at the conclusion that she wants to commercially exploit it [via entrepreneurship].” Diversity of thinking is associated with diversity in various kinds of demographic, social, and economic backgrounds, such as gender, race, place of origin, education, and income. Measuring diversity based on countries of origin, Audretsch et al. (2010) and Qian (2013) reported a positive association between diversity and knowledge spillover entrepreneurship in German and U.S. regions.

Finally, certain types of *culture* have demonstrated to be important facilitators of (or barriers to) knowledge spillover entrepreneurship. Culture is a multidimensional concept. In the literature on KSTE or innovation-driven spin-offs, some cultural factors, including willingness to collaborate, openness, hierarchy, social capital, and the organizational culture of universities, have been discussed. For instance, Saxenian (1994) notably compared the high-technology industries in Silicon Valley and Route 128 in the Boston area. According to her findings, cultural factors that encourage information exchanges, knowledge sharing, and high mobility in the labor market—including collaboration, openness, and lack of hierarchy—make Silicon Valley a more vibrant entrepreneurial community. These regional characteristics similarly play an important role in knowledge spillover entrepreneurship that depends on information exchanges, knowledge sharing, and the mobility of employees (to become entrepreneurs). Related to collaboration and openness discussed by Saxenian (1994), social capital, or simply trust, among community members facilitates interpersonal interaction. Apparently, one is more likely to collaborate or share information with someone else whom he/she trusts. Indeed, trust and interaction reinforce each other. As Malecki (1994) pointed out, trust “is largely, if not entirely, based on past experience” (p. 137). Increased interaction among researchers, entrepreneurs, and other community members in the regional entrepreneurial ecosystem creates more opportunities for knowledge spillover entrepreneurship. Empirically, Landry, Amara, and Rherrad (2006) found a positive effect of university researchers’ social capital on their probability of starting spin-offs.

In a regional economy with the presence of one or more research universities, the potential of entrepreneurial activity as one way to commercialize university knowledge is highly hinged on the organizational culture of universities. Empirical studies (Motoyama & Bell-Masterson, 2014; Qian, 2017) have shown that the presence of research universities is not significantly associated with regional technology entrepreneurship in U.S. metropolitan areas. Hayter (2013) even found a negative relationship between post-spin-off university assistance and the performance of academic spin-offs. Although a few universities are well known for their support for faculty, student, and alumni entrepreneurship, represented by The Massachusetts Institute of Technology (MIT) and Stanford University, many research universities have not been successful in realizing their entrepreneurial potential. Qian and Yao (2017) have studied two college-town entrepreneurial ecosystems: Boulder, Colorado, and Iowa City, Iowa. Boulder, home to the University of Colorado–Boulder, is the most entrepreneurial American college town measured by the start-up rate. In contrast, Iowa City, home to the University of Iowa, is a typical Midwest college town with low start-up activity. Qian and Yao (2017) found that, despite facing vastly different entrepreneurial environments, these two major public research universities present similar cultural barriers to faculty, student, and alumni entrepreneurship, including bureaucratic slowness, risk aversion, unsupportive attitudes toward faculty start-ups, and “silo” university structure. Strengthening knowledge spillover entrepreneurship from university spin-offs calls for changes in the organizational culture of universities.

Based on what has been discussed, Figure 1 provides a schematic framework to understand knowledge spillover entrepreneurship in regional economic development. Differentiated knowledge bases represent differentiated entrepreneurial opportunities, and therefore lead to regional variations in knowledge spillover entrepreneurship. The extent to which knowledge can be commercialized by entrepreneurs depends on entrepreneurial absorptive capacity and environmental factors in the regional entrepreneurial ecosystem, notably including competition, networks of people, diversity, and culture. Human capital is critically

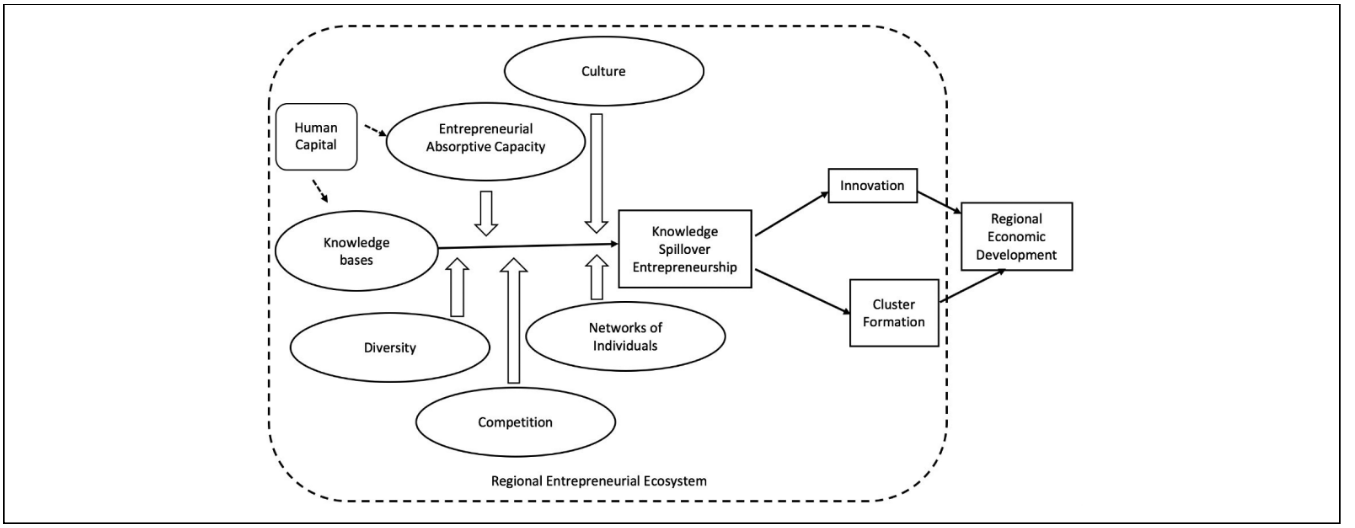

Figure 1. Regional entrepreneurial ecosystems for knowledge spillover entrepreneurship.

important, as it is the key determinant of both knowledge bases and entrepreneurial absorptive capacity (Qian & Acs, 2013). It is worth mentioning that these environmental factors may interact with each other, a key characteristic of the regional entrepreneurial ecosystem. For instance, networks may influence both competition and culture, which will be explained in the concluding section. A vibrant regional entrepreneurial ecosystem gives rise to knowledge commercialization (i.e., innovations) and cluster formation, ultimately leading to long-term economic development.

Concluding Remarks: Economic Development Policy Facilitating Knowledge Spillover Entrepreneurship

This research reviews the literature at the intersection of knowledge spillovers, entrepreneurship, and regional economic development. The key arguments include new knowledge does not necessarily give rise to regional growth; entrepreneurship can serve as a mechanism of geographically mediated knowledge spillovers, which leads to accelerated knowledge commercialization; knowledge spillover entrepreneurship also contributes to regional development by means of cluster formation; and knowledge spillover entrepreneurship is dependent on or moderated by historical, social, and economic factors in the regional entrepreneurial ecosystem, notably including knowledge bases, absorptive capacity, competition, networks, diversity, and culture.

Are there economic development policies that *particularly* boost knowledge spillover entrepreneurship? From the finance perspective, one U.S. federal program, the Small Business Innovation Research (SBIR) grant, has been acclaimed for encouraging technology start-ups (Audretsch, Weigand, & Weigand, 2002; Qian & Haynes, 2014). This program supports small- and medium-sized enterprises (SMEs) for their innovative efforts. While not all technology start-ups are examples of knowledge spillover entrepreneurship, Toole and Czarnitzki (2007) specifically have shown that SBIR provides incentives for U.S. biomedical scientists to create new firms and commercialize their research output, providing direct evidence supporting the SBIR program's effect on knowledge spillover entrepreneurship from universities. SBIR is the most prominent competitive grant for small business innovation efforts as a federal program that could offer an SME up to $1.15 million in two phases. Recognition from receiving SBIR grants also helps SMEs to further secure venture capital (Lerner, 1996).

At the state level, the varying state laws regarding the enforceability of covenants not to compete have stood out in the context of knowledge spillover entrepreneurship or innovation-driven spin-offs (Franco & Mitchell, 2008; Gilson, 1999; Samila & Sorenson, 2011). These covenants allow employers to protect themselves from potential competition created by departing employees in either other competing firms or start-ups. Under such covenants, departing employees typically promise not to work in the same industry for a certain period (commonly 1 or 2 years). In the KSTE framework, the flexibility of employees to leave their employers and start a new firm in the same industry is apparently critical. Strong enforcement of noncompete covenants threatens such flexibility and therefore discourages knowledge spillover entrepreneurship or technology spin-offs (Gilson, 1999; Samila & Sorenson, 2011). Employees who want to start their own firms are further discouraged by the challenges to hire experienced workers who are also subject to covenants not to compete (Samila & Sorenson, 2011). Contrary to the focus of Saxenian (1994) on cultural factors, Gilson (1999) and Franco and Mitchell (2008) attributed Silicon Valley's overtaking of Route 128 as the U.S. high technology center to the legal infrastructure related to the enforcement of non-compete covenants: For historical reasons, California does not enforce them while Massachusetts does. Samila and Sorenson (2011) have clearly suggested that states adopt more employee-friendly labor laws so that their efforts to build high-technology clusters can be more effective.

At the regional or local levels, Feld (2012) has argued that entrepreneurs should be the "leader," while support organizations, such as universities and local governments, should play a "feeding" role in regional entrepreneurial ecosystems. Indeed, the leadership of successful entrepreneurs characterizes the most successful entrepreneurial ecosystems including Silicon Valley and Boulder. How can local governments best feed an entrepreneurial ecosystem for vibrant knowledge spillover entrepreneurship? Based on the discussion from the previous section, this review ends with a highlight of two local government efforts that are highly desired: addressing the need for networks, which may also benefit a healthy competition environment and cultural changes, and the need for diverse talent or human capital.

Networks, as noted earlier, are critical to knowledge spillovers and the coordination of various resources needed for technology start-ups. Fortunately, local governments do not incur much financial burden to strengthen local networks. For instance, they can serve as conveners among technology communities, sponsors of local entrepreneurship events (such as small meet-up events and mega entrepreneurship celebration events), or ad hoc connectors of various stakeholders. These networking opportunities facilitated by the government help knowledge sharing in the entrepreneurial ecosystem and resource matching for start-up activity, therefore directly accelerating knowledge spillover entrepreneurship. As one example of the interconnectedness among factors in the regional entrepreneurial ecosystem, enhanced networks may also affect two additional factors framed in Figure 1: competition and culture. When networking opportunities better connect entrepreneurs with local incumbent firms, the latter group will be less likely to deter the start-up activity of the former one, creating a healthy competition environment. Moreover, continuous government efforts toward strengthening networks also help build trust among community members (Malecki, 1994) and change the local culture hostile to start-ups when the public perceives the government as a cheerleader of entrepreneurship due to their visible networking efforts.

It is perhaps more difficult to address the human capital need; however, human capital is essential to knowledge spillover entrepreneurship as it shapes the knowledge base of entrepreneurial ecosystems (Asheim & Coenen, 2005) and determines regional entrepreneurial absorptive capacity (Qian & Acs, 2013). Additionally, high diversity in the backgrounds of human capital makes knowledge spillover entrepreneurship more likely to occur (Audretsch et al., 2010; Qian, 2013). To address the need for human capital, one direction of public efforts is to invest in local schools, community colleges, and universities. This is particularly important for small cities that cannot compete with large innovation hubs (e.g., Silicon Valley, Boston, or New York) but have indigenous advantage to retain their "homegrown" talent. Qian and Yao (2017) reported that most of the recent star entrepreneurs in Iowa City are graduates of the University of Iowa, the anchor institution of this college town. These talented entrepreneurs do not move to the coastal areas, to a large extent, because they are native Iowans and/or proud "Hawkeyes" (nickname of the University of Iowa). In most cases, however, home ties are not sufficient. Talent is highly mobile, and indeed its attraction and retention have been a challenge to many cities or regions. Both the labor market demand and quality of life/amenities have impacts on the locational decision making of talent (Roback, 1982). Moreover, the tolerance of cities plays an important role in attracting talent with various backgrounds (Florida, 2014; Qian et al., 2013). The efforts of local governments in these aspects depend on the specific situation of each region.

The difficulties in providing specific policy guidelines for talent attraction and retention reflect one limitation in the current state of the KSTE research: Overall, policy discussion based on KSTE is too generalized and often difficult to operationalize in economic development practice. This, in part, arises from the heterogeneity of regional economies that requires different policies and plans in different regions even for the same goal, such as talent attraction. However, it would promote the real-world impacts of the theory if KSTE scholars and economic development practitioners from some representative regions work together toward more practically applicable plans under different scenarios.

Declaration of Conflicting Interests

The author declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author received no financial support for the research, authorship, and/or publication of this article.

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